Engineering Mechanics 1

Week 11

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Autonomous Aerial Systems

Potential Energy

- **Conservative Force**: Generated work depends on starting and end point, not on the course.
- **Potential Energy**: Measure of the ability to perform work (when a conservative force acts on a body)
- Examples:

  - **Weight force → Gravitational energy:** $V_g(h) = mg(h - h_0)$

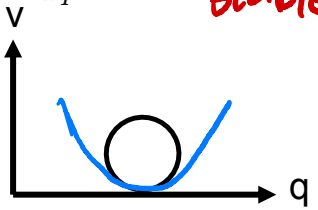
  

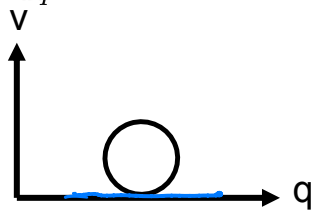
  - **Spring force → Elastic potential:** $V_e(x) = \frac{1}{2}k(x - x_0)^2$

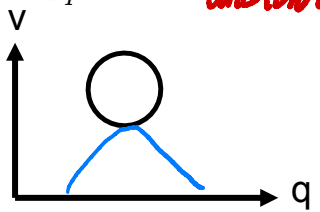
  - **Torsion spring moment → Elastic potential:** $V_e(\varphi) = \frac{1}{2}c(\varphi - \varphi_0)^2$

- **Equilibrium**: $\frac{\mathrm{d}V(q)}{\mathrm{d}q} = 0$ $\quad \delta W = F \cdot \delta q = \frac{dV(q)}{dq} \cdot \delta q$

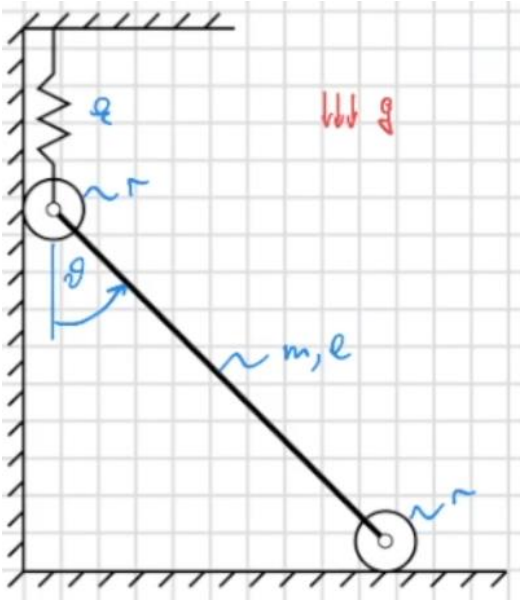
- **Stability**: $\frac{\mathrm{d}^2V(q)}{\mathrm{d}q^2} > 0$ stable $\quad \frac{\mathrm{d}^2V(q)}{\mathrm{d}q^2} > 0$ neutral $\quad \frac{\mathrm{d}^2V(q)}{\mathrm{d}q^2} > 0$ unstable







Exercise 1



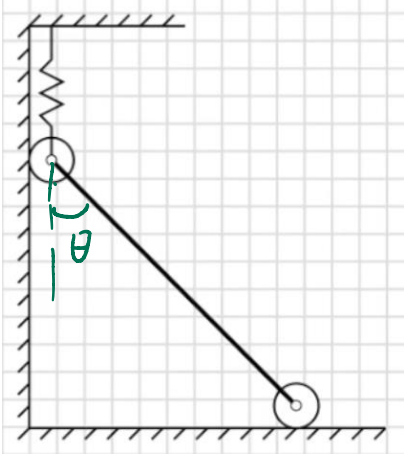
We know

- Gravity, g = 9.81 m/s2
- Spring coeff, k = 200 N/m
- Beam, m = 10kg, l = 0.6m
- Wheels, massless, r = 0m (approx.)
- All is frictionless
- Spring relaxed for theta = 0

Calculate

- Angle theta for equilibrium
- Evaluation of equilibrium

Potential Energy.

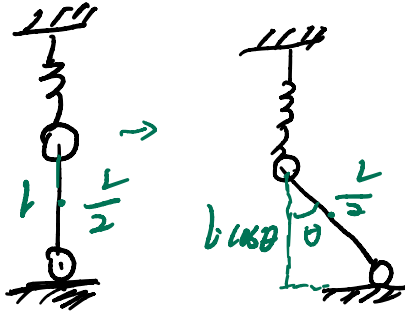


GP + EP.

$V(\theta) = V_g(\theta) + V_e(\theta)$

$= mg\frac{1}{2}(l\cos\theta - l) + \frac{1}{2}k(l - l\cos\theta)^2$

Equilibrium $\frac{dV(\theta)}{d\theta} = 0$.



$\frac{dV(\theta)}{d\theta} = -\frac{1}{2}mgl\sin\theta + kl^2(1-\cos\theta)\sin\theta$

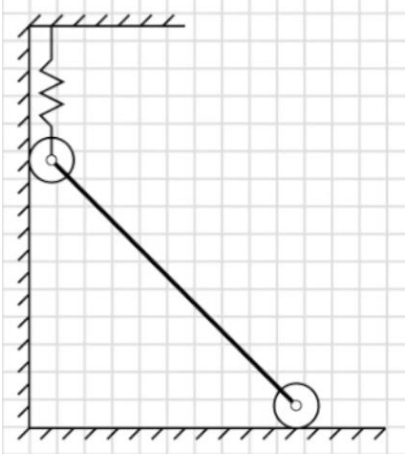
$= \sin\theta\left(-\frac{1}{2}mg + kl^2 - kl^2\cos\theta\right)$

Δ

$\frac{dV(\theta)}{d\theta} = 0 \Rightarrow \begin{cases} \sin\theta = 0 \\ \Delta = 0 \end{cases} \Rightarrow \begin{cases} \theta_1 = 0 \\ \theta_2 = \arccos\left(1 - \frac{mg}{2kl}\right) = 53.8^\circ \end{cases}$

Stability

$$\frac{d^2V(\theta)}{d\theta^2} = -\frac{1}{2}mgl\cos\theta + kl^2[\sin^2\theta + (1-\cos\theta)\cos\theta]$$



① $\theta_1 = 0$.

$$\left.\frac{d^2V(\theta)}{d\theta^2}\right|_{\theta=\theta_1} = -\frac{1}{2}mgl < 0.$$

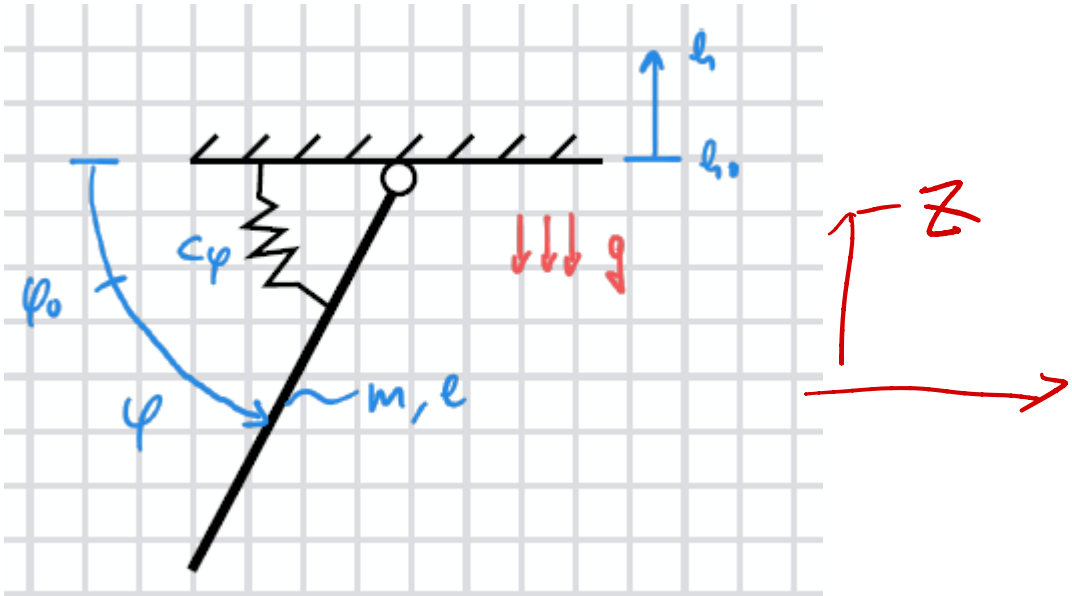
unstable equilibrium point.

② $\theta_2 = 53.8^\circ$

$$\left.\frac{dV(\theta)}{d\theta}\right|_{\theta=\theta_2} = 46.9127 > 0$$

stable equilibrium point

Exercise 2



We know

- m, l, g, c_φ
- $\varphi \in (0, \frac{\pi}{2})$

Calculate

- Potential
- Existence of equilibrium
- Stability analysis

Potential E.

$$U(\varphi) = V_g(\varphi) + V_e(\varphi)$$

$$= -\frac{1}{2}mgL\sin\varphi + \frac{1}{2}C_\varphi(\varphi-\varphi_0)^2$$

$-\sin(\varphi-\varphi_0)$

$$\frac{dU(\varphi)}{d\varphi} = -\frac{1}{2}mgL\cos\varphi + C_\varphi(\varphi-\varphi_0) = 0$$

$$\varphi \in [0, \tfrac{\pi}{2}]$$

$$\left.\frac{dU(\varphi)}{d\varphi}\right|_{\varphi=0} = -\frac{1}{2}mgL - C_\varphi\varphi_0 < 0,$$

$$\left.\frac{dU(\varphi)}{d\varphi}\right|_{\varphi=\frac{\pi}{2}} = C_\varphi(\tfrac{\pi}{2}-\varphi_0) > 0,$$

$$\exists\ \varphi \in (0, \tfrac{\pi}{2}) \Rightarrow \left.\frac{dU(\varphi)}{d\varphi}\right|_{\varphi} = 0.$$

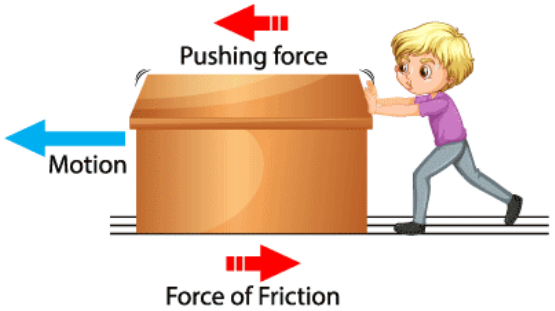
$\Rightarrow \quad \varphi \in (0, \frac{\pi}{2})$

$$\frac{d^2 V(\varphi)}{d^2 \varphi} = \frac{1}{2} mgl \sin\varphi + C_\varphi \quad > 0$$

$\Rightarrow$ the equilibrium point is stable.

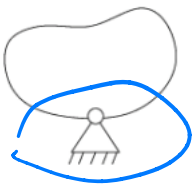
Friction



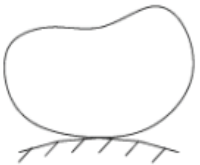


1. **Dry fiction** (Coulomb friction):

Ideal rough
(no relative motion)



General body contact



Ideal smooth
(no friction)



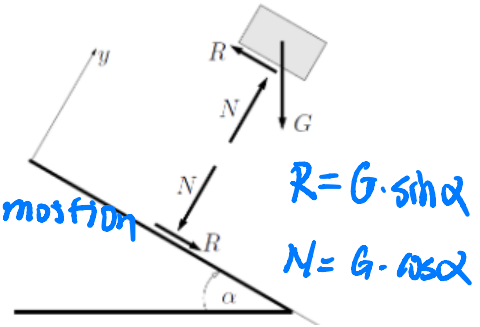
Friction

Coefficient of static friction: u_0 depending on materials rubbing against each other

***Static** friction*

- Adhesion: $|\vec{R}| \leq u_0 \cdot |\vec{N}|$
- Friction angle: α
- Properties: ① direction: opposit potential motion direction

② $|\vec{R}| \leq u_0 |\vec{N}|$



***Edge** case*

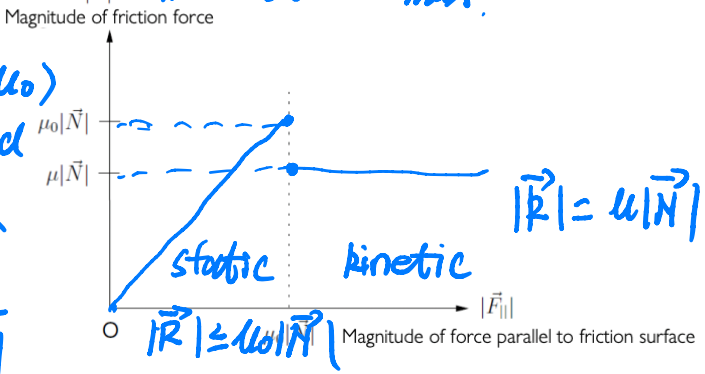
- Adhesion limit: $|\vec{R}| = u_0 |\vec{N}| \Rightarrow \sin\alpha_{max} = u_0 \cdot \cos\alpha_{max}$
- Maximum tilting angle: $\alpha_{max} = \arctan(u_0)$

***Kinetic** friction* ⇒ work is performed

- Sliding: $\alpha > \alpha_{max}$, $R < G\sin\alpha$
- Properties:

① direction: $\vec{R} = -|\vec{R}| \frac{\vec{v}_{rel}}{|\vec{v}_{rel}|}$

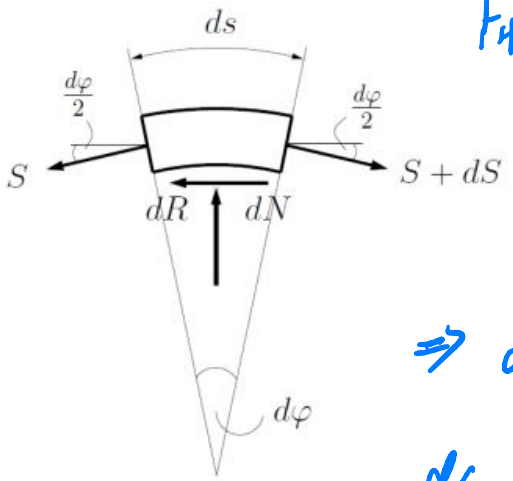
② magnitude: $|\vec{R}| = u \cdot |\vec{N}| \Rightarrow u < u_0$



Friction

static system: Equilibrium

2. Belt friction (Euler –Eytelwein):



$$F_V = S \cdot \sin\frac{d\varphi}{2} + (S + dS)\sin\frac{d\varphi}{2} - dN = 0$$

$$F_H = -S \cdot \cos\frac{d\varphi}{2} + (S + dS)\cos\frac{d\varphi}{2} - dR = 0$$

$$dR = \mu_0 \cdot dN$$

$$\Rightarrow \sin\frac{d\varphi}{2} = \frac{d\varphi}{2}$$

$$\Rightarrow dS = \mu_0 \left(S\,d\varphi + \underbrace{\frac{1}{2} dS\,d\varphi}_{\to 0} \right)$$

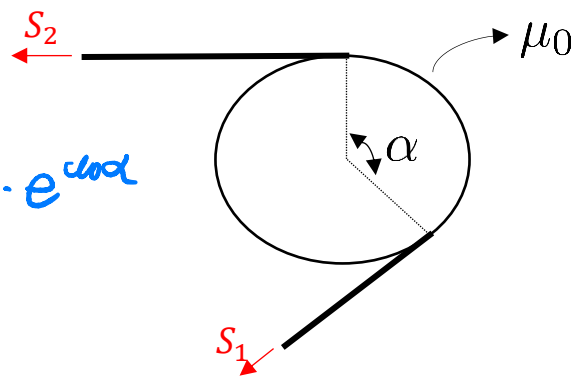
$$\Rightarrow \frac{dS}{S} = \mu_0 \, d\varphi \quad \text{(initial condition: } S(\varphi = 0) = S_2\text{)}$$

$$\Rightarrow \text{integration}: \quad \ln\frac{S(\varphi)}{S_2} = \mu_0 (\varphi - \varphi_0) = \mu_0 \varphi$$

Friction

2. **Belt friction** (Euler –Eytelwein):

for $\varphi = \alpha$, with $S_1 > S_2$



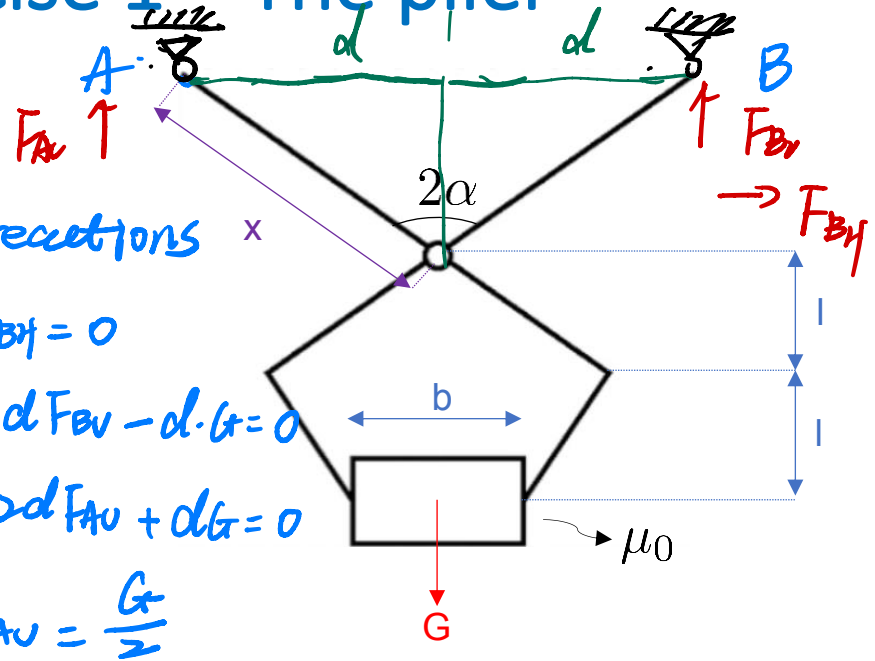
static edge case: $S_1 = S_2 \cdot e^{\mu_0 \alpha}$

$\Rightarrow$ static if $S_1 < S_2 \cdot e^{\mu_0 \alpha}$

Consider $S_1 < S_2$

static: $S_2 \cdot e^{-\mu_0 \alpha}$.

Exercise 1 – The plier



1> Support reactions

$\Sigma F_H = 0$, $F_{BH} = 0$

$\Sigma M_A = 0$, $2d F_{BV} - d \cdot G = 0$

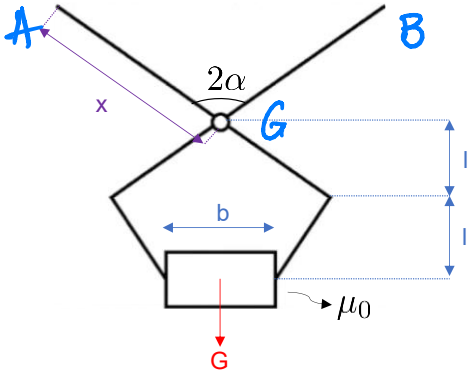
$\Sigma M_B = 0$, $-2d F_{AV} + dG = 0$

$\Rightarrow F_{BV} = F_{AV} = \frac{G}{2}$

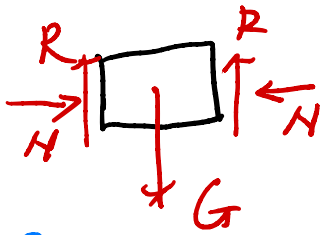
We know
- Weight of the object → G
- Distances → l, b
- Angle → 2α
- Static Friction coeff → μ_0

Calculate
- x, so that the object does not slip the plier

1) free cut.

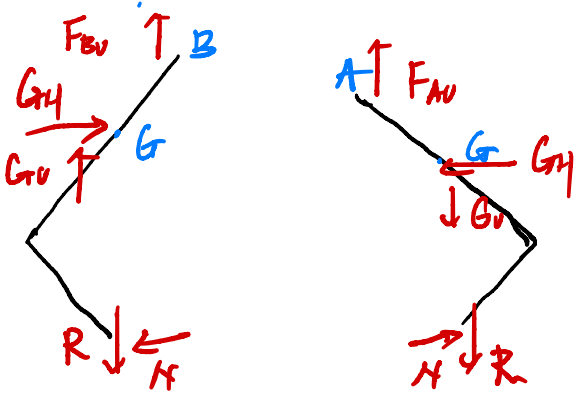


① load



$R = \frac{G}{2}$. $R = \mu_0 N$. $N = \frac{R}{\mu_0} = \frac{G}{2\mu_0}$

② left/right arm



③ left arm

$\Sigma F_V = 0$, $F_{BV} + G_V - R = 0$

$G_V = 0$

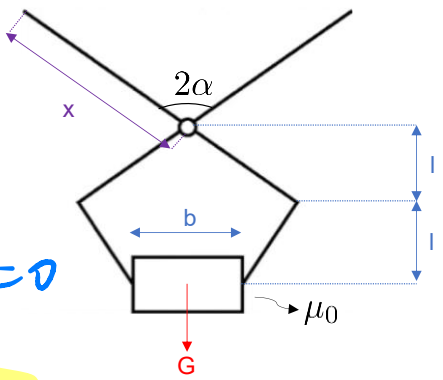
$\Sigma F_H = 0$, $G_H - N = 0$

$G_H = \frac{G}{2\mu_0}$

$M_G = 0$, $F_{AV} \cdot x \cdot \sin\alpha + R \cdot \frac{b}{2} - N \cdot 2l = 0$

$\frac{G}{2} \cdot x \cdot \sin\alpha + \frac{G}{2} \cdot \frac{b}{2} - \frac{G}{2\mu_0} \cdot 2l = 0$

$$x = \frac{1}{\sin\alpha}\left(\frac{2l}{\mu_0} - \frac{b}{2}\right)$$



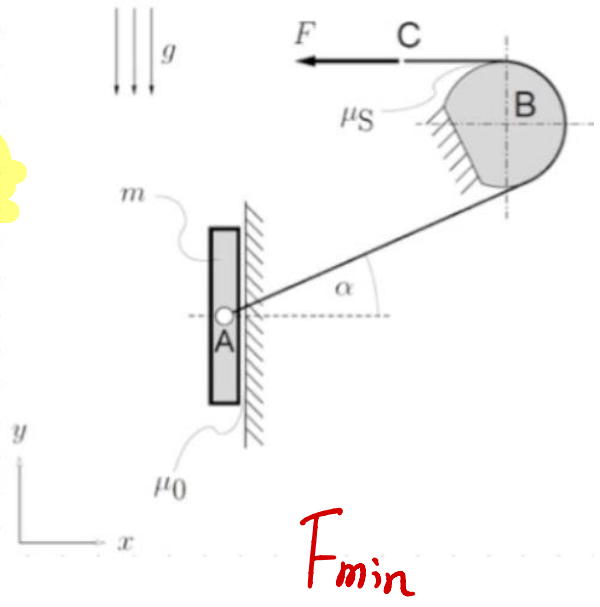
Exercise 2 – Rope friction

Question:

Calculate the minimum necessary value of the force applied at point C so that the plate just does not slip in the negative y-direction.

We know:

- Coordinate system→ x,y
- Mass and gravity→ m, g
- Friction coefficients → μ0, μs
- Angle interval → α in [0,90] grad



Free cut

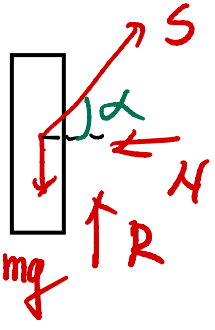
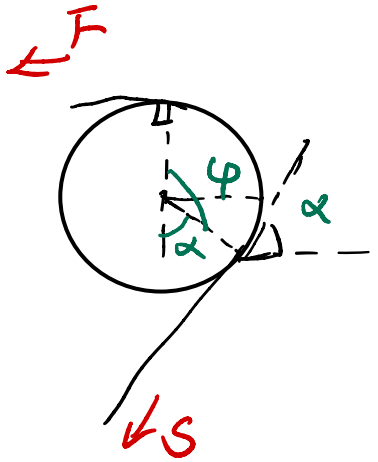


plate:



$\varphi = \pi - \alpha$

$$\Sigma F_V = 0 \Rightarrow R - mg + S\cdot\sin\alpha = 0$$

$$\Rightarrow R = \mu_0 \cdot N$$

$$\Sigma F_H = 0 \Rightarrow S\cos\alpha - N = 0$$

$$\Rightarrow \mu_0 \cdot S\cos\alpha + S\sin\alpha = G \Rightarrow S = \frac{mg}{\mu_0\cos\alpha + \sin\alpha}$$

Belt friction: $S\cdot e^{-\mu_0\alpha} \le F \le S\cdot e^{\mu_0\alpha}$

$$F_{min} = S\cdot e^{-\mu_0\varphi} = \frac{mg}{\mu_0\cos\alpha + \sin\alpha} e^{-\mu_0(\pi-\alpha)}$$